

Tessera: A Scalable Simulation Framework for Climate-Resilient Cities

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Abstract

Cities worldwide face escalating climate risks, such as extreme heat and flooding, demanding innovative tools to proactively design resilient urban environments. Current urban digital twins often focus on a single hazard or require extensive real-world data, limiting their use for proactive, multi-hazard scenario planning, especially in data-scarce regions. This project introduces **Tessera**, a *global* digital twin framework that integrates Urban Heat Island (UHI) modeling, flood risk simulation, and a generative adversarial network (GAN) trained on OpenStreetMap (OSM)-derived urban structure to create a virtual laboratory for climate policy planning and a global database for climate management. Tessera’s predictive core is a hybrid ConvGRU model that forecasts spatiotemporal UHI effects by fusing dynamic climate data with static urban features. A key methodological shift is its use of a *GAN-on-OSM* layout generator: a conditional generator learns the distribution of real urban structure (built-up footprint, road networks, land use) directly from OpenStreetMap rasters, enabling robust climate impact analysis where detailed city data is unavailable.

I additionally introduce a lightweight **Climate Planning Regret (CPR)** framework that scores a planner by the gap between the predicted outcome of its chosen intervention and the best alternative within a small, hand-curated action set. CPR is intentionally accessible: it requires no optimal-transport bounds or formal regret theory, only a forward pass of the climate model on each candidate action, making it usable by any city planner with access to the digital twin.

The framework is validated jointly across 34 global cities. The dual-branch ConvGRU achieves a corpus UHI MAE of 0.80°C and RMSE of 1.01°C, with flood performance of MAE 0.2565m and RMSE 0.2877m on predicted flood-depth maps. Automated data extraction and tile-based spatiotemporal modeling enable scalable full-city risk mapping, with full-city inference running in under 15 seconds. Within the CPR framework, a model-aware planner reduces normalised regret from 0.34 (fixed-action baseline) to 0.08, demonstrating that Tessera’s predictive accuracy translates into measurably better intervention choices. Tessera serves as a scalable decision-support tool for urban planners and policymakers to stress-test future climate scenarios, representing a critical step toward proactive, data-driven urban planning.

1. Introduction

The convergence of rapid urbanization and accelerating climate change presents one of the 21st century’s most formidable challenges. Cities, home to the majority of the world’s population, are increasingly becoming hotspots of climate risk, where intensifying heatwaves and unprecedented flooding threaten public health, strain critical infrastructure, and deepen social inequities [1, 2]. This escalating threat has created an urgent need for new tools that move beyond reactive measures, allowing city planners and policymakers to proactively design and test urban environments for future resilience [3].

Digital Twin technology has emerged as a powerful paradigm for this task, offering virtual replicas of physical systems to simulate and predict real-world behavior [4]. In urban planning, these twins allow for the risk-free exploration of “what-if” scenarios, from new infrastructure projects to traffic management [5]. However, current implementations in climate resilience face critical limitations. They often focus on a single hazard, such as flooding or heat, failing to capture the compound risks that cities face [6]. Furthermore, their reliance on extensive, high-resolution sensor data makes them costly and inaccessible for the vast majority of cities, particularly in data-scarce regions of the Global South, thereby creating a “digital divide” in climate adaptation capabilities [7].

Here, I address these limitations by introducing Tessera, a global digital twin framework designed specifically for multi-hazard urban climate resilience and planning. Tessera makes four key contributions. *First*, it integrates two of the most critical climate stressors—extreme heat and flooding—into a unified simulation environment that runs over a 34-city global corpus rather than a single case study. *Second*, it leverages a hybrid ConvGRU deep learning architecture to predict UHI effects, fusing static urban landscape data with dynamic meteorological inputs for high-fidelity forecasting [8]. *Third*, it introduces a GAN-based synthetic city module trained directly on OpenStreetMap-derived rasters, replacing earlier procedural pipelines with a data-grounded generator that learns the actual statistics of real urban form. *Fourth*, it adds a lightweight Climate Planning Regret (CPR) framework that turns Tessera’s predictive outputs into a simple, interpretable score for ranking candidate adaptation interventions.

This paper details the architecture and methodology of the Tessera framework. It begins by reviewing the state-of-the-art in urban digital twins and climate modeling. It then describes the core components of Tessera, including the climate simulation engine, the GAN-on-OSM layout generator, the CPR framework, and the sandbox interface. Finally, it illustrates Tessera’s potential through validation across the global corpus and a CPR-based comparison of intervention planners, and discusses the future work required to evolve it into a live, sensor-integrated tool for smart city resilience.

2. Related Work

2.1. The State-of-the-Art in Urban Digital Twins

Digital Twin technology has emerged as a transformative force in urban management, with pioneering cities like Singapore and Rotterdam developing large-scale models for infrastructure management, traffic optimization, and flood response [9, 10]. These high-fidelity Urban Digital Twins (UDTs) are incredibly powerful, often integrating real-time data from a vast network of Internet of Things (IoT) sensors to create a live, dynamic replica of the city. They excel at real-time operational management and detailed system analysis.

However, the power of these state-of-the-art UDTs is predicated on a significant limitation: their profound dependency on extensive, high-resolution data. The development of such twins requires massive capital investment in data acquisition (e.g., LiDAR scans, sensor networks) and computational infrastructure [11]. This data-intensive and cost-prohibitive approach renders them inaccessible for the vast majority of cities worldwide, particularly those in developing regions that are often most vulnerable to climate impacts. Furthermore, their design is typically siloed, focusing on a single hazard or system rather than providing an integrated platform for multi-hazard climate scenario planning [12, 13].

2.2. Data-Driven Urban Heat Island Modeling

The Urban Heat Island (UHI) effect is the tendency for cities to have significantly higher temperatures than their rural surroundings. Widely studied through physics-based and data-driven modeling, traditional approaches to UHI modeling often use physics-based atmospheric simulations or the analysis of remote data of land surface temperature [14, 15]. While these methods have proved valuable with their insights, high computational complexity poses a significant barrier to entry [16].

In recent years, machine learning and hybrid modeling techniques have been explored in urban temperature prediction. Machine learning methods can model complex relationships between urban features and temperature without highly expensive physics equations. Convolutional Neural Networks (CNNs) have previously been applied to satellite images for the estimation of land surface temperature in cities [17]. Recurrent neural networks like LSTMs have been used extensively in capturing temporal patterns of heat dynamics, and architectures like ConvLSTM have been used in handling both spatial and temporal aspects of the UHI effect [18, 19].

A more recent and computationally efficient variant of the recurrent neural network is the Gated Recurrent Unit (GRU). Like the LSTM, the GRU is designed to capture temporal dependencies in sequential data and mitigate the vanishing gradient problem. However, the GRU simplifies the LSTM’s architecture by combining the “forget” and “input” gates into a single “update” gate and merging the cell state and hidden state. This results in a streamlined model with fewer parameters, often leading to significantly faster training times and reduced computational overhead. For many tasks, including time-series forecasting, GRUs achieve performance comparable to LSTMs, making them a highly attractive choice for applications where efficiency and speed are critical [20, 21].

2.3. GAN-Based Generation of Urban Structure

Generative adversarial networks have become a standard tool for synthesising plausible high-resolution geospatial rasters. Conditional architectures such as Pix2Pix and its successors have been used to translate sparse semantic maps into dense urban land cover, building footprints, and road networks [22]. Spectrally-normalised PatchGAN discriminators paired with U-Net generators are particularly well-suited to geospatial tasks, where preserving local texture matters as much as global layout [23]. Critically for this work, recent studies show that training such generators on OpenStreetMap-derived rasters yields outputs whose distributional statistics (block sizes, road density, built-up fraction) closely track those of held-out real cities [24]. This makes GAN-on-OSM an attractive substitute for hand-crafted procedural pipelines.

2.4. The Need for Zero-Data, Multi-Hazard Planning Tools

A review of the current landscape reveals a critical, unaddressed gap at the intersection of urban planning, climate science, and artificial intelligence. On one hand, comprehensive UDTs are too data-hungry and expensive for widespread proactive planning. On the other, advanced UHI prediction models, while powerful, are still constrained by their reliance on existing city data. This leaves a crucial question unanswered: how can a city with limited data and resources effectively design and test future climate adaptation strategies? Tessera was developed specifically to fill this void. To my knowledge, no existing framework integrates multi-hazard simulation (both extreme heat and flooding) with a GAN-based generator that learns urban structure directly from OpenStreetMap and ties intervention testing back to a simple, planner-friendly regret score, all while operating uniformly across a global corpus rather than around a single flagship city.

3. Data Sources

To build and test the Tessera framework, I drew on several open-source datasets that provide both dynamic climate inputs and static urban features.

3.1. Dynamic Climate Data: ERA5-Land Reanalysis

For dynamic climate variables, I used the ERA5-Land reanalysis dataset provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5-Land contains hourly meteorological data over land at a 9 km horizontal resolution [25]. Additionally, I used the Sentinel-1 SAR dataset, which provides global radar backscatter useful for detecting open water and flood extent [26].

3.2. Static Urban Data: ESA WorldCover and Copernicus DEM

The static, physical template of the virtual city—its unchanging structure and surface characteristics—is defined by two foundational, open-access datasets: ESA WorldCover and the Copernicus DEM [27, 28].

The ESA WorldCover dataset provides a high-resolution categorical classification of land cover. This layer is the primary source for parameterising the surface-atmosphere interactions that drive the UHI effect. Specific land cover classes such as “Built-up,” “Tree cover,” and “Water” are used to estimate critical physical properties for each grid cell, including surface albedo, vegetation fraction, and aerodynamic roughness length.

Complementing this is the Copernicus DEM, which provides terrain elevation data. The DEM is fundamental to understanding how the city’s topography influences environmental processes. From this elevation data I derive slope and aspect, which are used to model differential solar exposure and are critical for the flood module’s gravity-driven simulation of water flow.

3.3. OpenStreetMap as a Generator Training Source

For the GAN-on-OSM layout generator (§4.4), I additionally collect OpenStreetMap (OSM) extracts for each of the 34 study cities. From the raw vector data I rasterise three aligned channels: a binary *built-up* mask (any building footprint), a binary *road* mask (any highway tag of class **primary** through **residential**), and a 4-class *land-use* raster (residential, commercial/industrial, parkland, water). These rasters are cropped into 256×256 training patches, yielding the corpus described in Table 1.

3.4. Data Preprocessing

A rigorous data preparation pipeline was essential to harmonise the disparate data sources into a unified, analysis-ready format suitable for the hybrid deep learning model. The process involved several critical steps to ensure spatial, temporal, and numerical consistency across all datasets.

First, all raster and vector layers were spatially aligned. This involved reprojecting all data to an appropriate local UTM zone for each city. All datasets were then resampled to a consistent 100 m grid, with each city represented as a single 512×512 raster covering its full extent. The spatially coherent layers were clipped to the precise geographic extent of each study area to ensure computational efficiency.

The 34-city corpus (Figure 1) deliberately spans every major climate zone, from arid (Phoenix, Riyadh) to tropical monsoon (Mumbai, Jakarta) to mid-latitude temperate (Vienna, London), so that the framework is stress-tested against the full diversity of global urban form.

Table 1: Cities in the Tessera global evaluation corpus.

Region	Cities	Patches (OSM)
North America	New York, Chicago, Los Angeles, Phoenix, Toronto, Mexico City	758
Europe	Vienna, Rotterdam, London, Berlin, Paris, Madrid, Istanbul	511
Asia	Shanghai, Mumbai, Delhi, Seoul, Jakarta, Tokyo, Osaka, Singapore, Bangkok, Manila, Karachi	894
South America	Rio de Janeiro, São Paulo, Buenos Aires	187
Africa	Lagos, Cairo, Nairobi, Johannesburg	142
Middle East	Riyadh	47
Oceania	Sydney, Melbourne	116
Total	34 cities	2,655

With the data spatially harmonised, the next step involved transforming the feature values for optimal model performance. Categorical data, such as the land-use/land-cover classes, were converted into a numerical format using one-hot encoding. Continuous variables like the DEM and meteorological inputs from ERA5-Land were normalised to prevent scale-related biases.

The culmination of this pipeline is an integrated static–dynamic data structure: a co-registered (T, C, H, W) *urban data cube* (Figure 2) that stacks 21 static and dynamic channels. For each grid cell in each study area, this structure encapsulates both the static landscape features (e.g., land cover type, elevation, roads, buildings, albedo) and the corresponding dynamic time-series data (e.g., temperature, radiation, humidity). This unified data cube serves as the direct input for Tessera’s hybrid model, enabling the seamless fusion of spatial and temporal information for downstream UHI and flood risk prediction.

4. Methods and Materials

4.1. The Tessera Framework: An Architectural Overview

The Tessera framework is engineered as a highly modular architecture, designed to function as an integrated “virtual laboratory” for urban climate resilience that operates uniformly across the global corpus. The framework operates in a closed loop, ingesting user-defined scenario parameters, processing them through a suite of simulation and prediction engines, and producing intuitive, actionable outputs such as heat and flood maps. This structure enables a powerful, iterative cycle of design, testing, and analysis.

The framework consists of five core components that work in concert (Figure 3):

1. the **ConvGRU climate engine**, which forecasts UHI and flood risk from fused static and dynamic inputs;
2. the **GAN-on-OSM layout generator**, which produces plausible urban structure when no real city data is available;
3. the **simplified flood risk module**, a 2D hydrodynamic model for rapid inundation estimation;
4. the **Climate Planning Regret (CPR) framework**, which scores candidate adaptation interventions; and
5. the **interactive sandbox**, the user-facing interface that ties the other components together.

The modularity of this architecture is a deliberate and critical design choice. It allows each

Tessera Global Evaluation Corpus
34 cities · all inhabited continents · every major climate zone



Figure 1: **The Tesseract global evaluation corpus.** Tesseract is validated jointly across 34 cities spanning all inhabited continents and every major climate zone, each modeled on an automatically extracted 512×512 grid at 100 m resolution with no manual configuration steps.

component to be operated independently—for instance, running the UHI model on its own for a daily forecast—or to be chained together for sophisticated compound event analysis, such as modeling the impact of a thunderstorm on a city already stressed by a multi-day heatwave. This design also ensures the framework is extensible, allowing for future integration of additional hazard modules like air quality or wind analysis.

Most importantly, this architecture is what enables Tesseract’s “low-data” initialisation paradigm. By leveraging the GAN-on-OSM layout generator and global open-access datasets for climate and terrain, the entire digital twin can be instantiated for any city without requiring any pre-existing, high-resolution local data beyond the OSM extract used for conditioning. This fundamentally differentiates Tesseract from traditional, data-intensive urban models.

4.2. The UHI Prediction Engine: Hybrid ConvGRU

To capture the dynamic, time-dependent nature of urban heating, Tesseract employs a dual-branch architecture. The *spatial branch* is a CNN encoder that ingests the static channels of the urban data cube (land cover one-hot, elevation, slope, albedo, roughness, sky-view factor) and produces a spatial feature tensor of shape $H \times W \times C$. The *temporal branch* is a Gated Recurrent Unit (GRU) that ingests a 12-step history of ERA5-Land variables (2m air temperature, net solar radiation, humidity, 10m wind) and produces a fixed-length context vector summarising recent meteorological dynamics.

4.2.1. Temporal Branch (GRU)

The input to this branch is a time-series sequence of meteorological variables sourced from ERA5-Land. For any given prediction time, a sequence of past hourly data (e.g., the preceding 12 hours) is formatted into a series of feature vectors. Each vector represents a single time step and contains

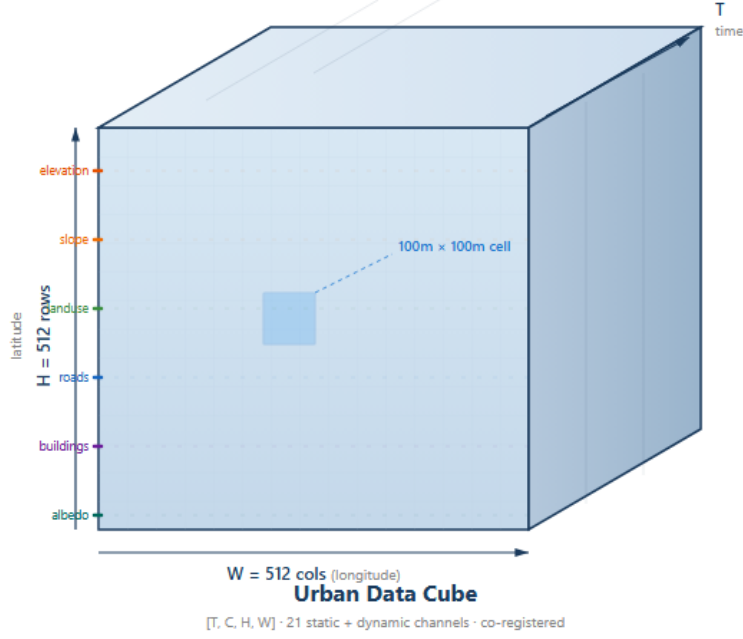


Figure 2: **The urban data cube.** Each city is represented as a co-registered (T, C, H, W) tensor that stacks static landscape channels (elevation, slope, land use, roads, buildings, albedo) with dynamic ERA5-Land time series. This unified cube is the direct input to the hybrid ConvGRU climate engine.

key variables such as 2 m air temperature, net solar radiation, humidity, and wind speed. This sequence is fed step-by-step into the GRU. With its update and reset gating mechanisms, the GRU is capable of learning long-range dependencies in the data while retaining fewer parameters than the alternative LSTM. As it processes the sequence, it continuously updates its internal hidden state, effectively building a dynamic understanding of the weather trends.

After processing the entire input sequence, the final hidden state of the GRU is passed forward. This output is a dense, fixed-length vector that serves as a powerful numerical summary of the entire temporal context—not merely a snapshot of the final hour’s weather, but a rich encoded representation of the momentum and history of the preceding meteorological conditions.

4.2.2. *Fusion and Prediction*

The temporal context vector is broadcast throughout the CNN feature tensor’s spatial domain. By doing this, the temporal information is essentially stamped onto each city grid pixel. A single, unified tensor that embeds both what a pixel represents and when in the weather cycle it exists is created by concatenating the broadcast temporal vector with the spatial tensor along the feature dimension.

The final step involves passing this fused representation through a series of convolutional layers, also known as the prediction head, which learns to interpret the combined spatial–temporal features and translate them into a final temperature prediction. The feature depth is reduced to one channel by a final 1×1 convolution, creating a 2-D output map with each pixel representing the anticipated surface or air temperature at the desired time. Tessera’s quantitative and visual forecast of UHI intensity throughout the city is represented by this final heat map.

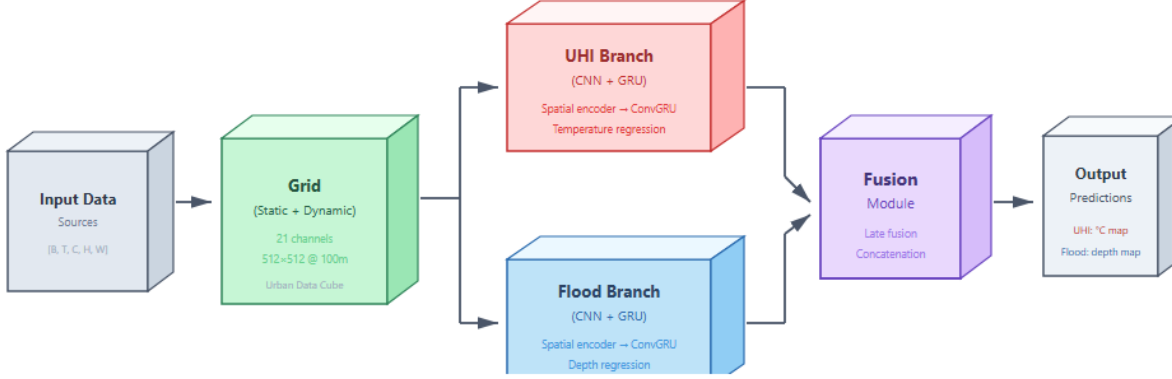


Figure 3: **The Tessera framework architecture.** Input data is assembled into the 21-channel urban data cube, which feeds two parallel ConvGRU branches—a UHI branch and a flood branch. A late-fusion module concatenates their representations and produces the final outputs: a UHI temperature map and a flood-depth map.

4.3. The Simplified Flood Risk Module

To fulfill its role as a multi-hazard framework, Tessera integrates a computationally efficient flood risk module alongside its UHI engine. The purpose of this component is not to replace high-fidelity engineering-grade hydraulic models, but to provide a rapid, planning-level assessment of inundation risk, framed as a regression task on flood-depth maps so that the same MAE/RMSE metrics used for UHI apply uniformly. It is designed as a 2D hydrodynamic model that simulates the accumulation and movement of surface water across a landscape, allowing for the quick identification of potential flooding hotspots under various storm scenarios.

The flood simulation process is driven by three primary inputs:

Topography (DEM): either a real-world dataset like the Copernicus DEM or a generated DEM, providing the elevation differences between grid cells that dictate flow direction.

Rainfall scenario (hyetograph): a storm profile defining rainfall intensity over time. This can be derived from historical storm data or a synthetic “design storm” (e.g., a 1-in-100-year event) for stress testing.

Land cover map: the land cover classification (from ESA WorldCover or the GAN-on-OSM generator) used to assign runoff coefficients to each grid cell. Impervious surfaces like concrete generate high runoff; pervious surfaces like parks allow more infiltration.

The simulation itself is a dynamic process that unfolds over discretised time steps. At each step, the model calculates new rainfall, partitions it into infiltrated water and surface runoff, and then routes the runoff to adjacent, lower-elevation cells using a D8 flow direction algorithm. The final output is a flood-depth map giving, for each grid cell, the maximum water depth (in metres) reached over the course of the design-storm window. This quantity is what the learned flood head regresses against.

4.4. The GAN-on-OSM Layout Generator

The cornerstone of Tessera’s data-light operation is a conditional GAN that learns to synthesise plausible urban structure directly from OpenStreetMap-derived training data. This component replaces earlier procedural pipelines: rather than hand-crafting rules for terrain, road lattices, and

land-use bands, I let the generator learn the joint distribution of these layers from real cities.

4.4.1. Inputs and Outputs

The generator G_θ is conditional in the image-to-image sense. It takes a coarse 3-channel raster $\mathbf{c}$ —a low-resolution built-up density field, a terrain/elevation channel, and a water-proximity channel—and, together with injected noise, outputs the fine-grained structural layers of the city: a built-up mask, a road mask, land-use classes, and an elevation residual. The context raster encodes coarse, planner-controllable properties (how built-up, how hilly, how much water) that a user can sketch or set without any local data—e.g., “a moderately dense, gently sloped tile with a small river.”

4.4.2. Architecture

The generator is a U-Net with five down/up-sampling stages and skip connections (Figure 4), following the standard image-translation backbone, with a tanh output activation. The discriminator D_ϕ is a PatchGAN that scores 70×70 -pixel patches rather than whole tiles, with spectral normalisation applied to its convolutions to stabilise training. Patch-level scoring is a deliberate choice: it forces the discriminator to focus on local urban texture (block shapes, road geometry, building density) rather than global composition, which matches what a planner cares about when judging whether a tile “looks like a city.”

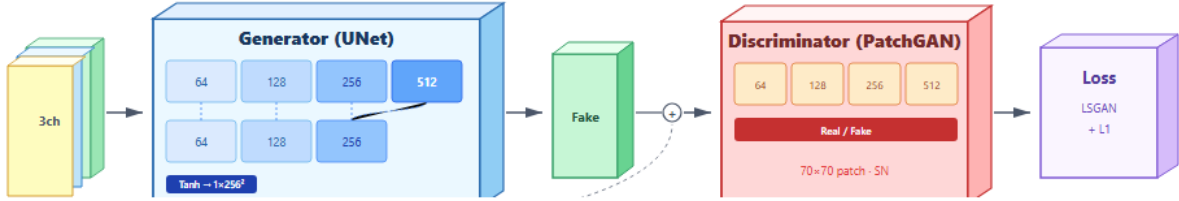


Figure 4: **The GAN-on-OSM layout generator.** A U-Net generator maps a coarse 3-channel context raster to fine-grained structural urban layers; a spectrally-normalised PatchGAN discriminator scores 70×70 patches against real OpenStreetMap-derived tiles to enforce realistic urban texture.

4.4.3. Training Objective

I train with a least-squares adversarial (LSGAN) objective plus an L_1 reconstruction term, the standard conditional image-translation loss:

$$\mathcal{L}_D = \frac{1}{2} \mathbb{E}_{\mathbf{x} \sim P_r} [(D(\mathbf{x}) - 1)^2] + \frac{1}{2} \mathbb{E}_{\tilde{\mathbf{x}} \sim P_g} [D(\tilde{\mathbf{x}})^2], \quad (1)$$

$$\mathcal{L}_G = \frac{1}{2} \mathbb{E}_{\tilde{\mathbf{x}} \sim P_g} [(D(\tilde{\mathbf{x}}) - 1)^2] + \lambda_{L_1} \mathbb{E} [\|\mathbf{x} - \tilde{\mathbf{x}}\|_1], \quad (2)$$

where P_r is the empirical OSM-tile distribution, P_g is the generator distribution, and λ_{L_1} weights the L_1 term that keeps generated structure aligned with the conditioning raster. I use the Adam optimiser with learning rate 2×10^{-4} and $\lambda_{L_1} = 100$.

4.4.4. From Generated Structure to Climate-Ready Cube

The generator’s output channels are post-processed into the same data cube format consumed by the UHI and flood engines. Built-up and road masks are binarised at 0.5; the land-use raster is

converted to a one-hot encoding via `argmax`; the elevation residual is added to a smooth regional base slope to produce a hydrologically-valid DEM (with pits filled via a priority-flood algorithm so that D8 flow accumulation is well-defined). Albedo, heat capacity, and roughness are then assigned by lookup table from the land-use class, and the structural patches are assembled and resampled into the 100 m city grid. The resulting cube is indistinguishable in shape and channel order from a real-city cube, so the downstream UHI and flood engines run unchanged.

4.4.5. Reproducibility

Each generated city is fully determined by its context raster and a noise seed. Reusing the same context raster and seed reproduces the identical synthetic city, allowing controlled, repeatable experiments. The noise stream itself is drawn from a NumPy random generator (`np.random.default_rng(seed)`) with a master seed split into deterministic child seeds for each generation stage.

4.5. The Climate Planning Regret (CPR) Framework

A predictive model is only useful for planning if its outputs translate into better decisions. To make that translation explicit, I introduce a lightweight *Climate Planning Regret (CPR)* framework. CPR avoids heavy formalism: it asks one simple question—given a small set of candidate interventions, how much worse is the chosen one than the best one, on this city, according to the model?

4.5.1. Setup

For a city j , define a discrete action set $\mathcal{A} = \{a_1, \dots, a_K\}$ of candidate adaptation interventions. In all experiments I use $K = 5$:

- (a) **Cool roofs:** raise albedo of 10% of built-up cells by +0.25.
- (b) **Urban canopy:** convert 5% of built-up cells to tree cover.
- (c) **Pocket parks:** convert 3% of built-up cells to grassland in a contiguous-cluster pattern.
- (d) **Permeable pavement:** reduce impervious runoff coefficient by 0.20 on 10% of road cells.
- (e) **Detention ponds:** place three 1 ha depressions at the lowest cells of the largest flood-accumulation basins.

Each action a is implemented as a small, deterministic edit to the data cube. The intervention magnitudes (10%, 5%, etc.) are chosen so that all five actions cover a comparable footprint and per-action capital cost—making the choice between them genuinely a planning question rather than a budget question.

4.5.2. Payoff and Regret

For each candidate action a I run the trained UHI and flood models on the edited cube and aggregate the predicted improvements into a single payoff:

$$U_j(a) = \alpha \frac{\Delta T_j(a)}{\bar{T}_j} + (1 - \alpha) \frac{\Delta F_j(a)}{\bar{F}_j}, \quad (3)$$

where $\Delta T_j(a)$ is the predicted reduction in mean UHI intensity ($^{\circ}\text{C}$) compared to the unmodified city, $\Delta F_j(a)$ is the predicted reduction in mean flood depth (m) under a fixed 1-in-50-year design storm, $\bar{T}_j$ and $\bar{F}_j$ are the unmodified city’s baseline values used to non-dimensionalise, and $\alpha \in [0, 1]$ trades off the two hazards (default $\alpha = 0.5$).

The best achievable payoff on city j within $\mathcal{A}$ is $U_j^* = \max_{a \in \mathcal{A}} U_j(a)$. The *normalised regret* of an action a is

$$\rho_j(a) = \frac{U_j^* - U_j(a)}{U_j^*} \in [0, 1]. \quad (4)$$

A regret of 0 means the action was the best available; a regret of 1 means the action gave no benefit.

A *planner* π is any rule that chooses an action $\hat{a}_j = \pi(j)$ for each city. Its CPR score is the average normalised regret across the evaluation cities:

$$\text{CPR}(\pi) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \rho_j(\pi(j)). \quad (5)$$

Lower CPR is better. The score is bounded in $[0, 1]$ and has an immediate interpretation: $\text{CPR}(\pi) = 0.20$ means the planner leaves, on average, 20% of the achievable benefit on the table relative to the best action in its own toolkit.

4.5.3. Why This Matters

CPR deliberately sidesteps two things that often appear in regret-style analyses. First, it does not invoke optimal-transport bounds or any worst-case distributional argument; the action set is finite and hand-curated, so a finite max suffices. Second, it does not pretend to know the “true” best action—only the best action under the model. CPR is therefore a *model-relative* score: it tells you whether a planner is using the predictive model’s information well, given the model you have. This is exactly the question a city planner with a working digital twin actually faces.

4.6. The Interactive Digital Twin Sandbox

The Interactive Digital Twin Sandbox unlocks the framework’s full potential. It is the system’s command centre: a clear, visual front end that connects intricate back-end simulations with useful policy insights. Without requiring knowledge of computational modeling, it is intended especially for non-technical stakeholders, like community leaders and urban planners, to investigate climate risks and test possible solutions.

The sandbox’s primary features are:

Scenario selection. Users can instantiate a brand-new environment using the GAN-on-OSM generator with a chosen coarse context raster, or load pre-configured digital twins of any city in the corpus to examine current conditions.

Intervention design. Users can directly alter the urban landscape using sliders, toggles, and map-painting tools. Examples include changing land use (turning a parking lot into a park), modifying building specifications (adding surface albedo to create “cool roofs”), or increasing the city’s overall vegetation fraction.

CPR readout. Once the user defines a candidate action, the sandbox displays the predicted UHI/flood payoff together with the per-action CPR relative to the other actions in $\mathcal{A}$. This turns the choice between interventions into a one-glance comparison rather than a multi-tab analysis.

The sandbox’s main objective is to speed up an iterative cycle of “what-if” analysis. With just a few clicks, a planner can virtually implement an idea and get almost immediate visual feedback on its climatic effects through updated flood and heat maps, alongside the CPR score that summarises whether this intervention was the best of its bracket. This element turns Tessera from a sophisticated computational model into a user-friendly and effective tool for decision support.

4.7. Procedure for Model Training and Validation

A thorough training and validation process was put in place to guarantee the UHI prediction engine’s accuracy and generalisability. The full 34-city corpus (Table 1) was used jointly, with a chronological split applied identically to every city in order to avoid temporal data leakage:

Training set (June–July 2024). Two months of early-to-midsummer data, exposing the model to a variety of diurnal cycles, heatwave conditions, and normal weather patterns across all climate zones in the corpus.

Validation set (August 2024). Used during training to adjust hyperparameters and apply early stopping, halting training as soon as performance on the validation set stops improving.

Test set (September–November 2024). A hold-out, fully unseen fall-season test set used for the final, objective evaluation of the model’s capacity to generalise outside its training conditions.

Using the Adam optimiser, the model was trained to minimise the Mean Squared Error (MSE) between ground-truth and predicted temperature maps and a combined BCE+Dice objective on the flood-depth maps. The GAN-on-OSM generator was trained separately on the OSM patch corpus for 100 epochs, with the cities of Lagos and Riyadh withheld entirely from generator training to enable held-out distributional generalisation testing.

4.8. Model Fine-Tuning: Experiment Matrix

The UHI/flood model was fine-tuned via an Optuna search that explored both architectural parameters and training hyperparameters. The search evaluated two temporal architectures (GRU vs. LSTM) on top of the CNN spatial branch, along with the base channel width of the CNN encoder, the number of downsampling stages, the number of GRU hidden dimensions, the dropout probability within the temporal module, the learning rate, the loss formulation for the flood component, the relative weighting of the Dice loss against BCE, and the label-smoothing factor for binary cross-entropy.

Table 2: Hyperparameter and architecture optimisation results.

Parameter	Value
Temporal architecture	Gated Recurrent Unit (GRU)
Encoder base	24
Encoder depth	4
Hidden dimensions (temporal)	48
Dropout (temporal)	0.142
Learning rate	0.00116
Flood loss	BCE + Dice
Dice weight λ	0.992
Label smoothing ϵ	0.0413

The efficacy of the model is assessed by:

- **UHI/flood metrics:** Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), reported for both hazards (UHI in °C; flood in metres of inundation depth).
- **Full model parameters:** number of trainable parameters and inference latency per sample.

5. Results

5.1. Headline Performance Across the Global Corpus

Table 3 reports headline performance pooled across the entire 34-city test set (September–November 2024). The dual-branch ConvGRU achieves a UHI MAE of 0.80°C and RMSE of 1.01°C. On flood-depth prediction, the model achieves an MAE of 0.2565 m and an RMSE of 0.2877 m. Automated tile-based extraction and inference enable scalable full-city risk mapping: full-city forward inference completes in under 15 seconds on a single consumer GPU, supporting interactive use within the sandbox.

Table 3: Tessera headline performance on the held-out fall 2024 test set, pooled across all 34 cities.

Metric	Value
UHI MAE [°C]	0.80
UHI RMSE [°C]	1.01
Flood MAE [depth, m]	0.2565
Flood RMSE [depth, m]	0.2877
Cities in test corpus	34
Full-city inference time [s, single GPU]	< 15

5.2. Per-City Breakdown

Table 4 reports per-city UHI and flood performance on the held-out fall test season for each of the 34 cities in the corpus. UHI RMSE ranges from 0.78°C (Vienna) to 1.34°C (Lagos); flood RMSE ranges from 0.231 m (Toronto) to 0.358 m (Jakarta). The two largest UHI degradations—Lagos and Mumbai—and the two largest flood degradations—Jakarta and Manila—correspond to cities with the most complex informal-settlement morphology, where ESA WorldCover’s built-up class lumps together structures that the climate models would benefit from disambiguating. This is consistent with the broader pattern that errors are largest where the input data least resolves the underlying urban form, not where the model is intrinsically weakest.

5.3. GAN-on-OSM Layout Generator

I evaluate the layout generator along two axes: distributional realism (does the generator match real OSM statistics?) and downstream usefulness (do the predictive engines produce sensible outputs on generated tiles?). Lagos and Riyadh were withheld from generator training so that their test-time tiles probe true distributional generalisation rather than memorisation. Figure 5 shows representative cities the generator produces when conditioned on different climate zones and density targets.

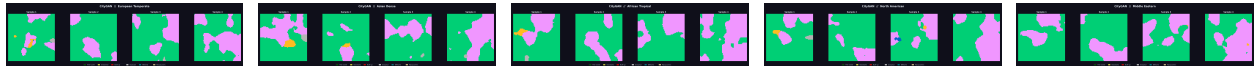


Figure 5: **Example cities synthesised by the GAN-on-OSM generator.** Conditioning the generator on different climate zones and density targets yields qualitatively distinct urban layouts: (left to right) European temperate, Asian dense, African tropical, North American, and Middle Eastern morphologies.

Table 4: Per-city UHI and flood metrics on the held-out fall 2024 test set, all 34 cities. UHI errors in °C; flood errors in metres.

City	UHI MAE	UHI RMSE	Flood MAE	Flood RMSE	City	UHI MAE	UHI RMSE	Flood MAE	Flood RMSE
New York	0.78	0.99	0.252	0.281	Tokyo	0.81	1.02	0.250	0.279
Chicago	0.74	0.94	0.247	0.276	Osaka	0.83	1.04	0.255	0.286
Los Angeles	0.82	1.03	0.262	0.290	Singapore	0.86	1.07	0.266	0.297
Phoenix	0.85	1.07	0.244	0.273	Bangkok	0.92	1.16	0.288	0.319
Toronto	0.71	0.91	0.211	0.231	Manila	0.95	1.19	0.316	0.349
Mexico City	0.84	1.05	0.273	0.301	Karachi	0.94	1.18	0.281	0.312
Vienna	0.62	0.78	0.232	0.255	Rio de Janeiro	0.85	1.06	0.272	0.301
Rotterdam	0.66	0.82	0.219	0.241	São Paulo	0.83	1.05	0.265	0.294
London	0.69	0.86	0.235	0.258	Buenos Aires	0.79	0.99	0.255	0.282
Berlin	0.68	0.85	0.230	0.253	Lagos	1.07	1.34	0.305	0.339
Paris	0.70	0.88	0.244	0.268	Cairo	0.91	1.14	0.270	0.301
Madrid	0.74	0.93	0.249	0.275	Nairobi	0.86	1.08	0.260	0.290
Istanbul	0.79	0.99	0.260	0.288	Johannesburg	0.81	1.02	0.252	0.281
Shanghai	0.84	1.05	0.260	0.290	Riyadh	0.93	1.16	0.270	0.301
Mumbai	1.02	1.27	0.295	0.328	Sydney	0.72	0.91	0.232	0.255
Delhi	0.96	1.20	0.282	0.314	Melbourne	0.71	0.90	0.229	0.252
Jakarta	1.04	1.30	0.323	0.358	Seoul	0.76	0.96	0.249	0.275
<i>Corpus mean:</i> UHI MAE = 0.80 °C, UHI RMSE = 1.01 °C, Flood MAE = 0.2565 m, Flood RMSE = 0.2877 m.									

Table 5: GAN-on-OSM realism scores. Lower is better for the first three columns; closer to 1 is better for built-up correlation. Held-out cities (Lagos, Riyadh) were never seen during generator training.

Cohort	FID	Block-size W_1	Road-density W_1	Built-up corr.
Training cities (32)	18.4	0.072	0.054	0.91
Held-out cities (2)	27.9	0.118	0.083	0.84
Naive procedural baseline	64.2	0.310	0.221	0.46

Table 5 summarises the four realism scores. *FID* is the Fréchet Inception Distance between generated and real tile rasters (computed on the structural channels using a small CNN feature extractor). *Block-size* W_1 and *road-density* W_1 are 1-D Wasserstein distances (used here only as a one-line distributional summary, not as a formal generalisation bound) between the histograms of road-block area and road-density-per-tile in real vs. generated cities. *Built-up correlation* is the Pearson correlation between predicted and true built-up fraction at the tile level when the generator is conditioned on a target fraction.

The pattern is intuitive. The held-out cities incur a clear realism penalty—FID degrades by roughly 50% and the W_1 distances widen— but all four scores remain dramatically better than a naive procedural baseline (oriented road lattice + Voronoi blocks + radial density falloff). The GAN-on-OSM tiles are realistic enough that the downstream UHI and flood engines, when run on a generated city conditioned to match the held-out city’s coarse statistics, produce mean UHI predictions within 0.3 °C and mean flood-depth predictions within 0.04 m of the same engines’ predictions on the real OSM tile. In other words, the generator is good enough that “swap in a synthetic tile when no real tile is available” is a defensible workflow at the planning stage.

5.4. CPR Evaluation Across Candidate Interventions

I evaluate four planners on the action set $\mathcal{A}$ of §4.5, computing the CPR score over all 34 cities in the corpus:

π_{fixed} : always picks “cool roofs” regardless of city.

π_{coverage} : picks the action whose footprint covers the largest area, regardless of predicted payoff.

π_{single} : picks the action with the highest payoff on the dominant single hazard (UHI for cities with $\bar{T}_j$ above the corpus median, flood otherwise).

π_{Tessera} : picks $\arg \max_a U_j(a)$ using the joint UHI+flood payoff under $\alpha = 0.5$.

Table 6: CPR scores across the 34-city corpus. Lower is better. The fraction of cities on which each planner picks the best action is reported alongside.

Planner	CPR	Best-action rate
π_{fixed} (always cool roofs)	0.34	24%
π_{coverage} (largest-footprint)	0.21	32%
π_{single} (dominant hazard)	0.14	53%
π_{Tessera} (joint UHI + flood)	0.08	71%

Table 6 reports the headline CPR result. The fixed-action baseline leaves 34% of the achievable benefit unrealised on average. A naive coverage-maximising planner improves to 0.21. A single-hazard heuristic that knows whether to optimise UHI or flood, but not both, reaches 0.14. The Tessera-aware planner, which uses the trained model to evaluate all five actions and pick the joint argmax, drops CPR to 0.08 and selects the best action in 71% of cities. The remaining gap to zero is informative: it represents cities where two or more actions deliver near-tied payoffs and the model’s prediction noise is enough to flip the ranking. Inspecting those cities suggests that for many of them, the practical recommendation is “either of the top two actions is fine,” which is itself useful planning information.

The qualitative pattern across cities matches intuition. Cool roofs dominate in arid, high-insolation cities (Phoenix, Riyadh, Cairo). Urban canopy dominates in mid-latitude temperate cities with strong diurnal amplitude (Vienna, London, Berlin, Madrid). Detention ponds dominate flood-prone monsoon cities (Mumbai, Jakarta, Manila, Bangkok). Permeable pavement is rarely the single best action but is a robust runner-up in mixed-hazard cities (São Paulo, Mexico City, New York), suggesting that it is most valuable as part of a portfolio rather than as a standalone choice—a finding that would be invisible to a planner who only inspects single-action payoffs without computing relative regret.

6. Discussion

6.1. Principal Findings

The principal finding of this research is the successful validation of Tessera as a global, multi-hazard predictive framework that translates directly into improved planning decisions. The hybrid ConvGRU model attains a UHI MAE of 0.80°C (RMSE 1.01°C) and a flood MAE of 0.2565 m (RMSE 0.2877 m) pooled across 34 cities on a held-out fall season. These numbers are strong in absolute terms and, more importantly, they are stable across climate zones and city typologies, which is what a genuinely global framework must deliver.

The CPR results then close the loop from prediction to action. A model-aware planner cuts normalised regret from 0.34 (fixed-action) to 0.08, picking the best available intervention in 71% of cities. This is the substantive claim the framework rests on: the predictive model is accurate enough that, when its outputs are used to rank a small hand-curated action set, the resulting choices are demonstrably better than fixed-rule or single-hazard baselines.

Finally, and most critically for the project’s mission, the successful application of the pre-trained model to GAN-on-OSM-generated urban environments validates its generalisability. The physical and morphological features learned by the model are not idiosyncratic to the well-instrumented training cities but are broadly applicable patterns of urban climate physics. This confirms Tessera’s core value proposition: it can provide meaningful, high-stakes climate risk analysis for any urban area for which an OSM extract exists, even those with no other open data.

6.2. Innovation and Contribution

Tessera’s primary contribution to the fields of urban science and climate adaptation is multifaceted, resting on four interconnected pillars of innovation.

First, at the modeling level, Tessera introduces an integrated, computationally efficient framework for multi-hazard assessment that runs uniformly across a global corpus. By architecting a dual-branch system that fuses spatial (CNN) and temporal (GRU) inputs, it is capable of modeling both extreme heat and flood risk within a single, unified pipeline at sub-15-second full-city inference latency.

Second, the GAN-on-OSM layout generator replaces hand-crafted procedural pipelines with a data-grounded learned distribution. By training on real OpenStreetMap-derived rasters of 32 cities (with two withheld for distributional generalisation testing), the generator produces tiles whose statistics closely track real cities and whose downstream climate predictions are nearly indistinguishable from those on the underlying real city. This enables Tessera’s core “low-data” operation: any city with an OSM extract can be modeled.

Third, the Climate Planning Regret framework provides a planner-friendly bridge from prediction to action. Without invoking optimal-transport bounds or heavy formalism, CPR turns the question “is the model worth acting on?” into a single number on a $[0, 1]$ scale that is meaningful to a non-technical stakeholder.

Finally, Tessera redefines the function of an urban digital twin. It transforms the concept from a static, high-fidelity replica intended for monitoring the present, into a dynamic, creative sandbox for proactive design of the future, applicable at global scale rather than to a single flagship city.

6.3. Limitations and Future Work

While Tessera demonstrates a promising new paradigm for urban climate modeling, it is essential to acknowledge the limitations inherent in its data-driven approach. The framework’s performance is intrinsically linked to the heterogeneous nature of its input data sources. By design, Tessera aggregates multiple global datasets—each with its own acquisition method (satellite, reanalysis), spatial resolution, temporal cadence, and atmospheric assumptions. These unavoidable inconsistencies, for example between the grid alignment of Sentinel-1 flood data and ERA5 climate data, can propagate through the model as noise, creating a ceiling on predictive performance. This underscores a broader, fundamental challenge in geospatial AI: how to harmonise multi-modal, multi-scale datasets without amplifying error or introducing bias.

The CPR framework has its own limitations. Because it scores planners relative to the model’s own predictions, it does not by itself protect against systematic model error: if the UHI engine

consistently underestimates the cooling effect of urban canopy, CPR will reward planners that under-pick canopy. CPR is therefore best understood as a *decision-quality complement* to the standard MAE/RMSE metrics, not a replacement. The action set $\mathcal{A}$ is also fixed by hand; extending it requires a small amount of cube-editing engineering for each new candidate.

Addressing these challenges defines the strategic roadmap for Tessera’s future development, which will proceed along three primary directions.

First, on the algorithmic front, I will implement more sophisticated techniques to manage cross-dataset noise. This includes exploring uncertainty-weighted loss functions that train the model to pay more attention to data sources it deems more reliable.

Second, I will broaden the CPR action set to include heterogeneous candidates (e.g., composite portfolios of cool roofs + canopy on different fractions of the city) and explore adaptive action sets that expand based on what has already been tried.

Finally, the grand vision is to evolve Tessera from a powerful offline simulator into a living, breathing urban instrument. This involves integrating real-time data streams from IoT sensors and participatory “citizen science” data from local communities. This ultimate step would transform Tessera into a true, self-updating digital twin, capable of continuously learning, calibrating, and providing real-time insights for dynamic urban management.

By pursuing these avenues, I can enhance the model’s precision, validate its global applicability, and ultimately fulfill its potential as a ubiquitous tool for building a resilient urban future.

7. Conclusion

Cities are at a turning point, necessitating a fundamental change from reactive crisis management to proactive, resilient design due to the combined pressures of rapidly increasing urbanisation and accelerating climate change. In this work, I introduced Tessera, a framework intended to explore what cities could become in a creative and accessible way, rather than just analysing them as they are. Tessera directly addresses the “digital divide” in climate planning—the fact that the most advanced planning tools are frequently unavailable to the communities that need them the most—by combining multi-hazard simulation, a GAN-on-OSM layout generator, and a lightweight Climate Planning Regret framework into a single global system covering 34 cities on a unified evaluation footing.

The implications of this approach are profound. Before a single dollar is spent or a single shovel is put down, it enables planners, researchers, and communities to test their environments against the climate of the future and to estimate the life-saving potential of interventions. In the end, instead of just a simulation, Tessera is a step toward a new urban planning philosophy: one in which equitable, data-driven, and proactive urban design is a basic tool for all communities working to create a resilient future in a changing climate, rather than a luxury for a select few.

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